

ch 7 > Risk Analysis



Source of Project Risk

- Cash flow Estimate
- Nature of Business
- Rate Of Return
- Study Period
- Social Risks
- Variability of O/P

Method of Describing Project Risk

A. Sensitivity Analysis

B. Break Even Volume

C. Scenario Analysis

A. Sensitivity Analysis

$$PW = -I + ANR \left(\frac{(1+i)^N - 1}{i(1+i)^N} \right) + \frac{S}{(1+i)^N}$$

$$PW = -I + ANR (P/A, i\%, N) + S(P/F, i\%, N)$$

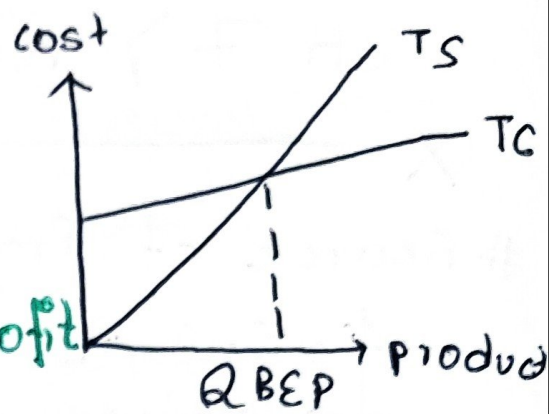
B. Break even Analysis

@ BE : Total sales = Total cost

$$S \times Q = FC + V \times Q$$

$$\# Q_{BEP} = \frac{FC + \text{Desired Profit}}{S - V}$$

(units)



$$\# Q = \frac{\text{Total Fixed cost}}{\text{Selling Price} - \text{Variable cost}}$$

C. Scenario (Prefeasibility) Scenario

- Done under 3 cases

Cases	Demand	Selling Price	variable cost
1. Worst	Low	Low	High
2. Average	Avg	Avg	Avg
3. Best	High	High	Low

following is given :

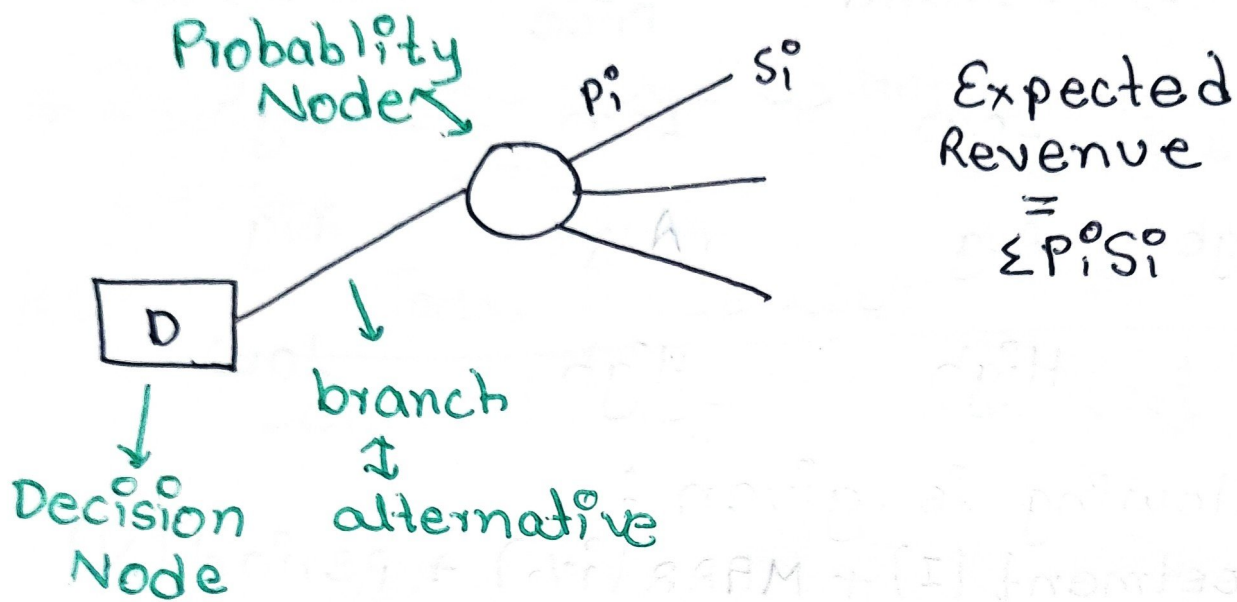
- Investment (I) + MARR ($i\%$) + period (N)
- Unit Demand (d)
- Unit Price (P)
- Variable cost per unit (VC) = $TC - FC$
- Fixed cost (FC)
- Salvage value (S)

→ Solⁿ

$$\begin{aligned} NPW = & -I \\ & + \\ & (d \times P - d \times VC - FC) (P/A, i\%, N) \\ & + \\ & S (P/F, i\%, N) \end{aligned}$$

Decision Tree

- Risk Tree Analysis w/ graphics



* chose Probability Node $\rightarrow$ w/ Higher Expected Revenue